

AARADHYA DEV TAMRAKAR

AI/ML · Edge AI · Computer Vision · Formal Methods · Robotics

Kathmandu, Nepal · aaradhyadevtmr@gmail.com · +977 9844602050 · [linkedin.com/in/aaradhya-dev-tamrakar](https://www.linkedin.com/in/aaradhya-dev-tamrakar) · github.com/AaradhyaDT · aaradhyadt.github.io

PROFILE

Final-year Electronics, Communication and Information Engineering student at IOE, Kathmandu Engineering College, focused on applied ML, edge AI, computer vision, statistical evaluation, and formal verification. Builds reproducible end-to-end systems spanning model training, quantitative evaluation, explainability, APIs, embedded inference, and SMT-based verification. Current work includes BiasAperture, SPARK, and a Headless Invariant Assurance Engine for state-machine verification and counterexample replay.

EDUCATION

Bachelor of Engineering (B.E.) Expected: January 2027
Electronics, Communication and Information Engineering — Tribhuvan University, Institute of Engineering (IOE), Kathmandu Engineering College | Year IV / Part II
Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems
Cumulative Percentage: 75%

EXPERIENCE

- Vice Chair**
IEEE KEC KTM Student Branch
 - Directed branch strategy, scaling initiatives and member engagement across executive committees.
 - Managed relations across the broader technical community.

2026 – Present
- Event Manager**
Electronics Project Club (EPC), Kathmandu Engineering College
 - Coordinated logistics for club workshops and department activities.

2026 – Present
- Makerspace Ambassador**
KEC Maker’s Space, Kathmandu Engineering College
 - Shipped site-wide fixes to the Maker’s Space website — image repair, content consistency, UI cleanup.

June 2026 – Present
- Vice Secretary**
IEEE KEC KTM Student Branch
 - Streamlined operations and grew program participation by managing documentation and orchestrating workshop logistics.
 - Coordinated volunteers alongside the committee for seamless execution.

2025 – 2026
- Resource Manager**
Electronics Project Club (EPC), Kathmandu Engineering College
 - Allocated scheduling and resources for workshops across multiple departments.

2024 – 2026

FELLOWSHIPS & RECOGNITIONS

- Fuse AI Fellow — Fusemachines AI Fellowship 2026**
Fusemachines
 - Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer), achieving 100% execution success and accuracy on a 50-question benchmark.
 - Built containerized AI services using FastAPI, PostgreSQL, and Docker.
 - Developed a churn/CLV pipeline achieving 5-fold CV ROC-AUC 0.841, with threshold tuning and SHAP explainability.
 - Applied Random Forest, XGBoost, Bayesian inference, Gaussian processes, clustering, and forecasting across applied datasets.
 - Worked with UCI Online Retail II (~500K transactions) for segmentation and forecasting, achieving MASE 2.44 with a four-model ensemble.

2026 – Present
- NSSR DataCamp Fellow — Cohort 2**
Nepalese Society of Student Researchers
 - Secured competitive selection (April 2026) for a sponsored DataCamp Premium license, one of a limited cohort.
 - Delivered six applied GenAI solutions, automating null filtering, deduplication, and schema normalization across product and data workflows.
 - Completed coursework toward SQL Associate and Python Data Associate certifications in PostgreSQL, Pandas, statistics, and scikit-learn.

2026 – Present

PROJECTS

BiasAperture — Vision Fairness & Bias Audit (Fellowship Capstone)

AIF360 · Fairlearn · FairFace · SHAP · Statistical Testing

- Developed a diagnostic fairness-auditing framework with dual AIF360/Fairlearn backends computing DPD, DIR, EOP, and EOD, with harmonized metric definitions and deterministic report generation.
- Validated on the FairFace benchmark (97,698 images) with BCa bootstrap confidence intervals ($B \geq 1,000$), χ^2 /Fisher testing, $n \geq 30$ safeguards, and Shapley attribution for flagged disparities.
- Co-authored with Tisha Manandhar; M1–M4 completed, M5 System Orchestration & Case Studies at 95%, with 78/78 repository tests passing.

SPARK — Wearable Fall Detection System (Major Project)

ESP32-S3 · 1D CNN · TensorFlow Lite Micro · SHAP · BLE · FastAPI

- Developed the ML pipeline on SisFall with subject-grouped 5-fold cross-validation, achieving 0.9185 AUC-ROC (38,420 windows) and 87.81% sensitivity; built an INT8 post-training quantizer exporting an 18.5 KB TFLite Micro model.
- Built the local BLE/serial gateway, SHAP attribution pipeline, JSON event store, PDF reporting, and REST dashboard; led ML modeling, quantization, and gateway work in a 4-person team.
- Repository test suite reports 48/48 passing; physical hardware bring-up, on-device BLE validation, and final deployment remain in the project roadmap.

Gesture-Controlled Self-Balancing Robot (GCSBR) — Minor Project

Computer Vision · Arduino · MPU6050 · Stepper Motors · Android · MATLAB

- Implemented PID stabilization firmware, validated via a MATLAB-simulation-to-hardware pipeline.
- Developed an Android gesture-control app (MediaPipe, CameraX, HC-05 BT) driving motors for real-time actuation.

Edge AI Stability Detection System — GCSBR Support Project

Python · scikit-learn · FastAPI · Joblib · IMU Data

- Predicted platform stability from simulated IMU data using a Random Forest model, reaching 99.8% test accuracy; served the persisted model via FastAPI with Joblib export for integration into GCSBR.

Headless Invariant Assurance Engine (INV-BMK-001)

Python · Z3 · SMT-LIB2 · State Machines · Sandbox Replay · Pytest

- Developed a formal verification benchmark that encodes state-machine contracts into SMT-LIB2/Z3, proves candidate invariants inductively, generates counterexamples, and replays them in a concrete Python sandbox; latest run evaluated 12 properties across 4 state machines, discovered and replay-confirmed 4/4 planted violations (100% recall), recorded 0.0% false-discovery rate, and measured 10.844 ms mean solver latency.

nepali-ocr-ai — Low-Resource Document Intelligence

Python · Multimodal Vision · Tesseract · Devanagari · OpenXML

- Prototyped a hybrid Nepali document pipeline combining multimodal vision OCR with local Tesseract fallback, Devanagari normalization, and contextual Varnavinyas checks; includes Preeti↔Unicode transcoding and DOCX repair workflows.

Super-NLM — Multi-Account NotebookLM Hub & MCP Server

Python · FastAPI · MCP · NotebookLM · asyncio

- Built an MCP server and web dashboard for multi-account NotebookLM access with round-robin routing, quota-aware cooldown/retry, multi-turn account pinning, cross-notebook synthesis, folder sync, and scheduled Studio artifact generation.

SKILLS

Programming: Python, C, C++, SQL, Kotlin, VHDL

ML/Statistical Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, PyMC, clustering, forecasting

Fairness/Explainability: SHAP, AIF360, Fairlearn, statistical testing, bootstrap confidence intervals

AI Systems: FastAPI, Docker, PostgreSQL, SQLite/FTS5, asyncio, MCP

Computer Vision/Edge AI: OpenCV, MediaPipe, TensorFlow Lite Micro, ONNX, Tesseract

Formal Methods/Verification: Z3, SMT-LIB2, state-machine verification, deterministic replay, pytest · Embedded/Robotics: ESP-IDF, FreeRTOS, ESP32-S3, Arduino, MPU6050, UART/I2C/SPI, MQTT, FPGA/Vivado · Tools: Git, GitHub, Jupyter, MATLAB, Colab, LaTeX/Overleaf

CERTIFICATIONS & ACTIVITIES

IEEE WIE Nepal LaTeX Workshop (May 2026) · IEEE SPaX organizing team · Prompt Engineering Fundamentals, TechAxis · IEEEExtreme 19.0 (Team ShadowXTREME) · PCB Design & Fabrication Workshop, KEC Robotics Club · AWS Cloud Computing Workshop · Microsoft Learn Student Ambassador

Active in IEEE KEC, EPC, KEC Maker's Space, and KEC Music Club. | Last Updated: September 30, 2026